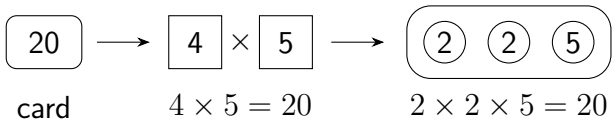
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 that cannot split. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Tokens with the same number are the same.

Problem 1: For each card, find two different splitting routes. Can different routes finish with different prime bags?

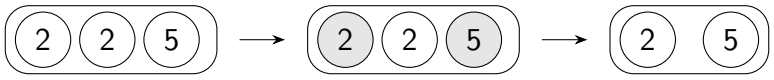
12

18

24

36

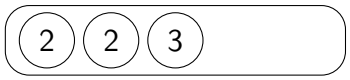
Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



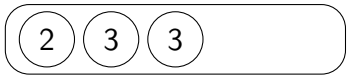
value 20 choose shaded tokens value 10

$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 2: For each bag, find every different value you can make by choosing tokens. Does your collection contain every divisor of the full bag's value?

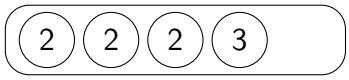


value 12

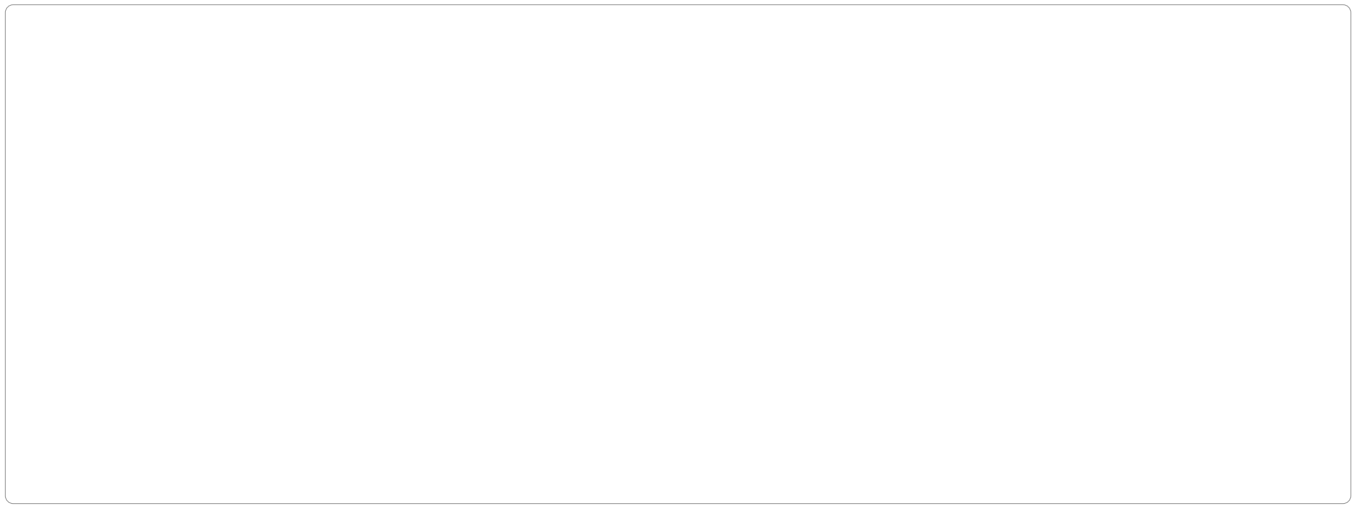


value 18

Problem 3: Find the complete collection for each bag. Does the bag with more tokens make more different values?

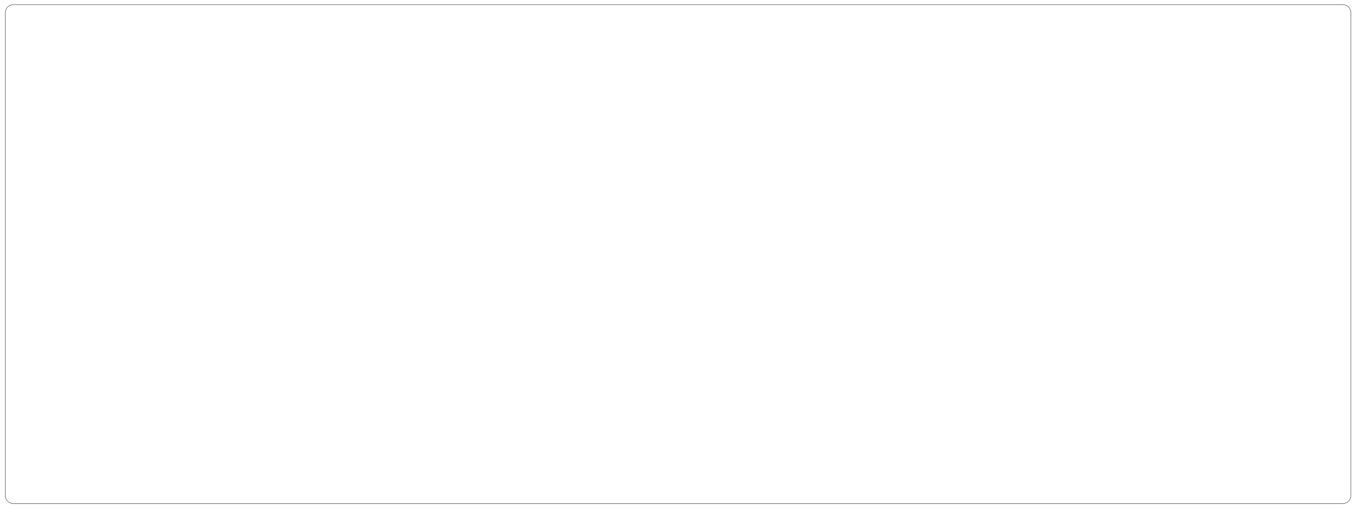


value 24





value 30



Problem 4: Use prime tokens with any of the labels 2, 3 and 5. Build a bag that makes exactly 4 different values by choosing tokens, another that makes exactly 6, and another that makes exactly 8. How small can you make the full bag's value in each case?

4 different values

6 different values

8 different values